KAIZEN MATHS

Lesson starter

Binomial Expansion

Specific Terms and Coefficients

Name _____

S1

Find the coefficient of x^2 in $(1-2x)^7$

S2

Find the 7th term in $(4-3x)^8$

S3

Find the coefficient of x^2 in $(1+3x)^7$

S4

Find the 5th term in $(3-2x)^8$

Show your working

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ALG-C4F7-L2 · S

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Exit ticket

Binomial Expansion

Specific Terms and Coefficients

Name

E1

Find the coefficient of x^4 in $(1+x)^6$

E2

Find the 6th term in $(5-2x)^7$

E3

Find the coefficient of x^5 in $(1+x)^9$

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ALG-C4F7-L2 · E

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Teacher answers & working

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S1

Find the coefficient of x^2 in $(1-2x)^7$

Answer:

84

Working:

Method focus: use the general term and match the required power.

$T_{r+1} = {}^7C_r(1)^5(-2x)^r$

${}^7C_2 = 21$

Coefficient $= 21(-2)^2 = 84$

Final answer: 84

S2

Find the 7th term in $(4-3x)^8$

Answer:

$326592x^6$

Working:

Method focus: the nth term uses r = n-1 in the general term.

The 7th term is T_7, so $r = 6$

$T_7 = {}^8C_6 4^2(-3x)^6$

$326592x^6$

Final answer: $326592x^6$

S3

Find the coefficient of x^2 in $(1+3x)^7$

Answer:

189

Working:

Method focus: use the general term and match the required power.

$T_{r+1} = {}^7C_r(1)^5(3x)^r$

${}^7C_2 = 21$

Coefficient $= 21(3)^2 = 189$

Final answer: 189

Teacher answers & working

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S4

Find the 5th term in $(3-2x)^8$

Answer:

$90720x^4$

Working:

Method focus: the nth term uses r = n-1 in the general term.

The 5th term is T_5, so $r=4$

$T_5 = {}^8C_4 3^4(-2x)^4$

$90720x^4$

Final answer: $90720x^4$

E1

Find the coefficient of x^4 in $(1+x)^6$

Answer:

15

Working:

Method focus: use the general term and match the required power.

General term

$T_{r+1} = {}^6C_r x^r$

Need $r=4$

${}^6C_4 = 15$

Final answer: 15

E2

Find the 6th term in $(5-2x)^7$

Answer:

$-16800x^5$

Working:

Method focus: the nth term uses r = n-1 in the general term.

The 6th term is T_6, so $r=5$

$T_6 = {}^7C_5 5^2(-2x)^5$

$-16800x^5$

Final answer: $-16800x^5$

Teacher answers & working

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E3

Find the coefficient of x^5 in $(1+x)^9$

Answer:

126

Working:

Method focus: use the general term and match the required power.

General term

$T_{r+1} = {}^9C_r x^r$

Need $r = 5$

${}^9C_5 = 126$

Final answer: 126